

Integrating factor

Key Points

To solve a 1st order differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Find the integrating factor

$$I = e^{\int P(x)dx}$$

Multiplying both sides by it then integrating and rearranging to solve for y

Remember the constant of integration

Basic Skills

Q1. For the differential equation $\frac{dy}{dx} = xy$

- a) Show that it is equivalent to $\int \frac{1}{y} dy = \int x dx$ and hence find the general solution to the differential equation
- b) Given that $y(2) = 1$ find the particular solution
- c) Rearrange the differential equation so that you can determine $P(x)$ and $Q(x)$
- d) By finding the integrating factor show that $\frac{d}{dx} \left(e^{-\frac{1}{2}x^2} y \right) = 0$
- e) Hence solve the equation for y and compare to your answer in part a)

Q2. Rearrange $x^2 \frac{dy}{dx} + e^x y + x = xy$ to determine $P(x)$ and $Q(x)$

Competency Skills

Q3. a) Verify that $y = 3e^{-5x} + 10x$ solves the differential equation $\frac{dy}{dx} + 5y = 10(1 + 5x)$

b) Find the general solution to the differential equation $\frac{dy}{dx} + 5y = 10(1 + 5x)$

Q4. a) Find the general solution to $\frac{dy}{dx} + \frac{2y}{x} = \frac{5}{x^3}$, verifying your solution for all values of your constant

b) Given $\frac{dy}{dx}(2) = 6$ find a particular solution

Q5. Find the particular solution to $\cos x \frac{dy}{dx} + y \sin x = e^x$ given that $y(0) = 1$

Q6. By using the substitution $u = \frac{y}{x}$ show that the differential equation $\frac{dy}{dx} - \frac{y}{x} = \frac{x}{y}$ can be written as $\frac{du}{dx} = \frac{1}{ux}$ and hence solve the differential equation using an appropriate method

Q7. By using the substitution $y = \frac{1}{u}$ find the general solution to $\frac{dy}{dx} - \frac{y}{x} = x^6 y^2$

Advanced Skills

Q8. Explain why $I \frac{dy}{dx} + IP(x)y = \frac{d}{dx} (Iy)$ in the integrating factor method

Q9. By using the substitution $u = \frac{1}{y}$ solve the differential equation $\frac{dy}{dx} + (y + x^2 y)^2 = \frac{y}{x}$

Q10. For the differential equation $(1-x) \frac{dy}{dx} + \frac{y}{1+x} = (1-x)^2$

- a) Find the general solution
- b) State the limitation implied by your solution
- c) Explain the behaviour of y as $x \rightarrow -1$

Extension

Q11. Solve for a general solution, the second order differential equation $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = 2x$

Hint: Consider how you could make the differential equation first order